

Experiment 2: Fine-Tuning Analysis

Probing the Attention-Router / MLP-Executor Division via LoRA Adaptation

1 Objective

We hypothesise that **MLP modules adapt more in magnitude and more specifically in direction** than attention modules upon domain-specific fine-tuning. Concretely:

1. MLP weight updates are *larger* (higher relative Frobenius norm) than attention updates.
2. MLP updates are more *domain-differentiated* (lower cross-domain cosine similarity) — different domains produce genuinely different MLP updates.
3. MLP updates are more *aligned with domain concept vectors* — the weight changes specifically modify the directions in representation space that carry domain identity.

This tests the storage/execution hypothesis from the adaptation side: if MLPs store domain-specific knowledge, fine-tuning should preferentially modify MLP weights in domain-specific directions.

2 Fine-Tuning Setup

We fine-tune six independent LoRA adapters on Qwen2.5-7B, one per arXiv domain, from the same base checkpoint. All adapters share identical hyperparameters:

Table 1: LoRA fine-tuning configuration.

Parameter	Value
Base model	Qwen/Qwen2.5-7B (bfloat16)
LoRA rank r	16
LoRA α	32
LoRA dropout	0.1
Target modules	q/k/v/o_proj, gate/up/down_proj
Training samples	3,000 per domain
Validation samples	1,000 per domain
Max sequence length	256 tokens
Epochs	3
Effective batch size	$4 \times 8 = 32$
Learning rate	1×10^{-4} (cosine schedule, 5% warmup)
Weight decay	0.01
Precision	bfloat16

2.1 Fine-Tuning Results

Figure 1 shows the train and validation perplexity across epochs for all six domains. All domains show clear training loss reduction from the baseline.

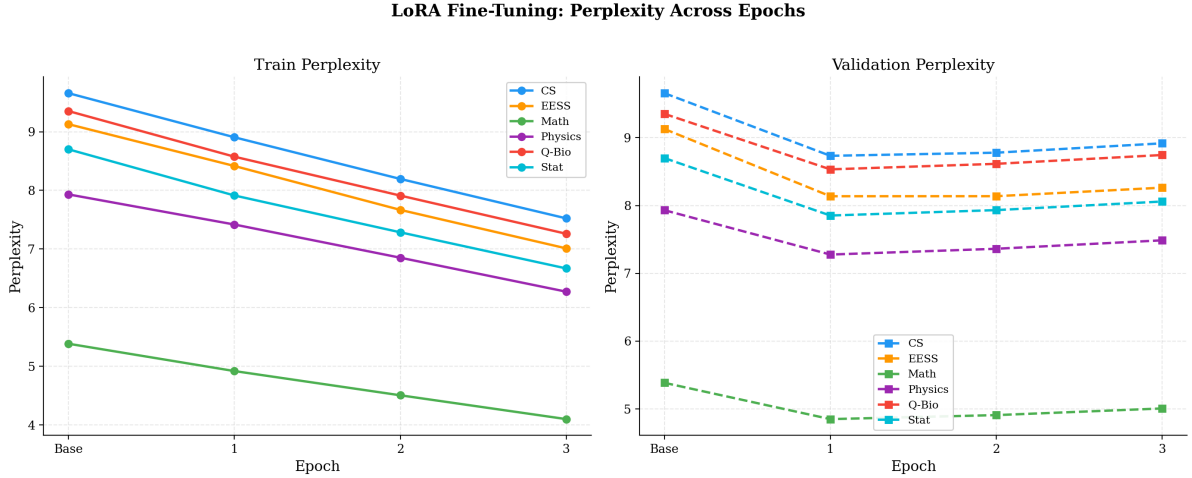


Figure 1: Train (left) and validation (right) perplexity across epochs. Epoch 0 denotes the base model baseline. All domains improve on training data; validation diverges after epoch 1.

Table 2: Perplexity across epochs. Best validation in **bold**.

Domain	Epoch 1		Epoch 2		Epoch 3		Baseline
	Train	Val	Train	Val	Train	Val	Val
CS	8.91	8.73	8.20	8.78	7.52	8.92	9.66
EESS	8.42	8.14	7.67	8.14	7.01	8.26	9.13
Math	4.92	4.85	4.51	4.91	4.10	5.01	5.39
Physics	7.42	7.28	6.85	7.36	6.27	7.49	7.93
Q-Bio	8.57	8.53	7.91	8.62	7.26	8.75	9.35
Stat	7.91	7.85	7.29	7.93	6.67	8.06	8.70

After exploring multiple configurations, we were unable to prevent overfitting due to the small dataset size (3,000 samples per domain). Validation perplexity begins increasing after epoch 1 for all domains while training perplexity continues to decrease. We therefore run all subsequent analyses independently across all three epochs to observe how the attention-MLP adaptation asymmetry evolves as training progresses and the model begins to overfit.

3 Experiment 2.1: Weight Change Norms

3.1 Hypothesis

MLP weight updates should be **larger in magnitude** than attention updates, as measured by relative Frobenius norm $\delta = \|\Delta W\|_F / \|W_{\text{base}}\|_F$. This would indicate that MLP modules undergo more substantial adaptation during domain-specific fine-tuning.

3.2 Method

For each domain k , epoch, layer ℓ , and module m , we compute the effective LoRA update $\Delta W = (\alpha/r) \cdot BA$ and measure:

$$\delta_m^{(k),\ell} = \frac{\|\Delta W_m^{(k),\ell}\|_F}{\|W_{m,\text{base}}^\ell\|_F}$$

We report both grouped (mean across attention modules vs. mean across MLP modules) and per-module breakdowns.

3.3 Results

Table 3: Mean relative weight change norm δ across all layers (Epoch 3). MLP consistently exceeds attention by $1.24\text{--}1.40\times$.

Domain	$\bar{\delta}_{\text{attn}}$	$\bar{\delta}_{\text{mlp}}$	Ratio
CS	0.00700	0.00913	$1.30\times$
EESS	0.00698	0.00976	$1.40\times$
Math	0.00696	0.00864	$1.24\times$
Physics	0.00697	0.00870	$1.25\times$
Q-Bio	0.00705	0.00920	$1.31\times$
Stat	0.00700	0.00905	$1.29\times$

MLP updates are consistently $1.24\text{--}1.40\times$ larger than attention updates across all domains. The effect is robust: *no domain* violates this pattern. The ratio is modest compared to prior work reporting $2\text{--}3\times$, which is attributable to the difference in base model (Qwen2.5-7B vs. Llama) and LoRA targeting all 7 modules rather than a subset.



Figure 2: Grouped weight change norms (δ_{attn} vs. δ_{mlp}) across layers for all six domains (Epoch 3). MLP (orange) consistently exceeds Attention (blue) at nearly every layer.

4 Experiment 2.2: Domain Specificity of Updates

4.1 Hypothesis

MLP weight updates should be more **domain-differentiated** than attention updates. When fine-tuning on physics, the MLP update should look different from when fine-tuning on math. Attention updates, by contrast, should be more similar across domains — attention adapts generically regardless of which domain is being learned. This directly addresses the concern that large MLP updates might just reflect a favourable optimisation landscape rather than domain-specific content.

4.2 Method

For each layer ℓ and module group (attention pooled, MLP pooled), we vectorise the weight updates from all six domains and compute pairwise cosine similarity:

$$\text{sim}(k, k')^\ell = \frac{\langle \text{vec}(\Delta W_\ell^{(k)}), \text{vec}(\Delta W_\ell^{(k')}) \rangle}{\|\text{vec}(\Delta W_\ell^{(k)})\| \cdot \|\text{vec}(\Delta W_\ell^{(k')})\|}$$

This produces a 6×6 cross-domain similarity matrix at each layer. We summarise this with two **domain mixing** scalar metrics, computed over off-diagonal elements:

- **MSE mixing score:** $\text{mean}(S_{\text{offdiag}}^2)$ — penalises large cross-domain similarity quadratically.
- **MAE mixing score:** $\text{mean}(|S_{\text{offdiag}}|)$ — linear average of cross-domain similarity.

Both range from 0 (perfectly domain-specialised, identity-like matrix) to 1 (fully domain-mixed, uniform matrix).

4.3 Results

The prediction is confirmed: MLP updates are consistently more domain-differentiated (lower mixing scores) than attention updates.

Table 4: Domain mixing scores — grand mean across all 28 layers (Epoch 3). Lower = more domain-specialised.

Metric	Attention	MLP
MSE ($\text{mean}(S_{\text{offdiag}}^2)$)	0.0275	0.0118
MAE ($\text{mean}(S_{\text{offdiag}})$)	0.1387	0.0880
Mean off-diagonal similarity	0.1387	0.0880

By the MSE metric, MLP is $2.3\times$ more domain-specialised than attention. By MAE, the ratio is $1.6\times$. The MSE metric amplifies the signal because attention has a few high cross-domain pairs (e.g. v_proj) that are penalised quadratically.

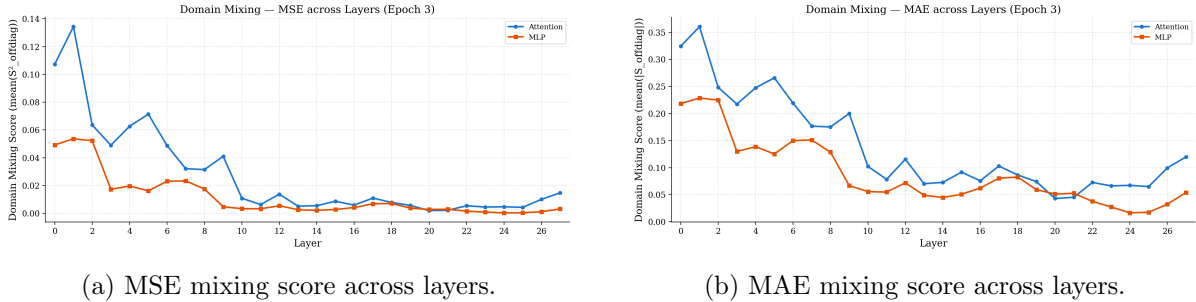


Figure 3: Domain mixing scores across layers (Epoch 3). Higher values indicate more domain-mixed (less specialised) updates. Attention (blue) consistently shows higher mixing than MLP (orange), especially in early layers (0–9).

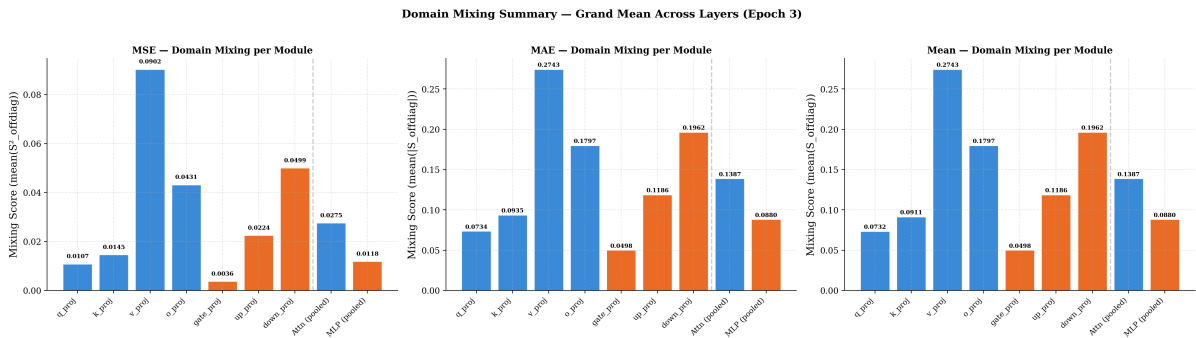


Figure 4: Per-module domain mixing summary (grand mean across all layers, Epoch 3). Blue = attention modules, orange = MLP modules. `gate_proj` is the most domain-specialised module overall (lowest mixing score). Among attention modules, `v_proj` is the most domain-mixed.

Module-wise drill-down. To identify which specific modules drive the attention–MLP gap, we examine the single most distinguishing layer — the layer where the off-diagonal similarity gap between attention and MLP is largest. For Epoch 3, this is **Layer 5** (attention off-diag mean = 0.266, MLP off-diag mean = 0.125, gap = 0.141).

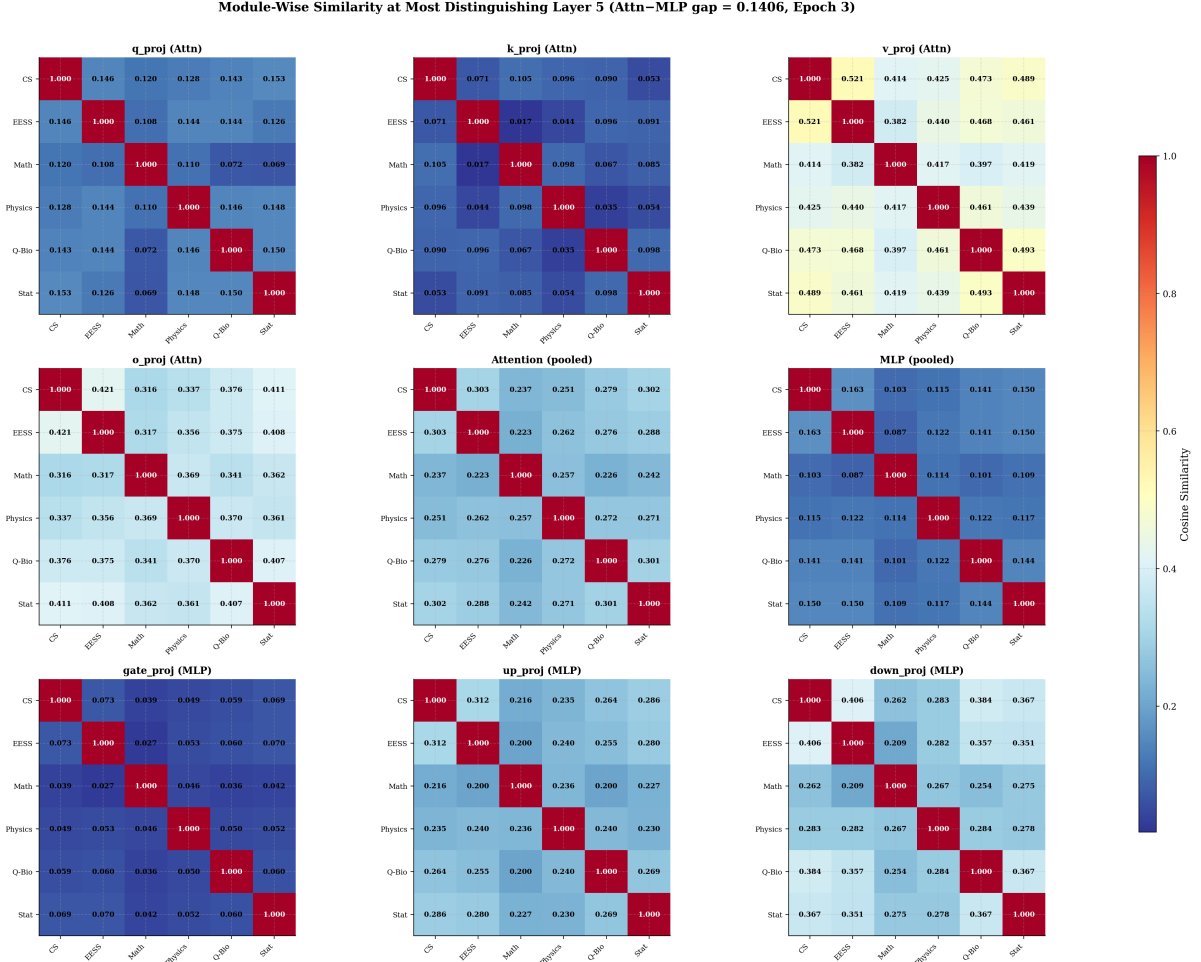


Figure 5: Module-wise 6×6 similarity heatmaps at Layer 5 (Epoch 3). Top row: attention modules (q/k/v_proj). Middle: o_proj and pooled groups. Bottom: MLP modules (gate/up/down_proj). All share a common colour scale. **gate_proj** (bottom-left) shows near-zero off-diagonal similarity — almost perfectly domain-specialised. **v_proj** (top-right) shows the highest cross-domain similarity within attention.

5 Experiment 2.3: Domain Alignment

5.1 Hypothesis

MLP weight updates should be more **aligned with domain concept vectors** than attention updates. The domain concept vectors $\hat{d}_k^\ell$ (computed from the forward pass geometry in Experiment 1) define the directions in representation space that carry domain identity. If MLP stores domain-specific knowledge, its weight updates should specifically modify these directions, rather than changing weights in orthogonal, domain-irrelevant directions.

5.2 Method

We measure domain alignment using two complementary metrics:

Projection metric (ρ). For each domain k , layer ℓ , and module m :

$$\rho_m^{(k),\ell} = \frac{\|\Delta W_m^{(k),\ell} \cdot \hat{d}_k^{\text{mlp},\ell}\|}{\|\Delta W_m^{(k),\ell}\|_F}$$

This measures: when the weight update is applied to the domain concept direction, how large is the output relative to the total update norm? High ρ means the update specifically operates on the domain-identity direction.

We pool across modules within each group (attention vs. MLP) using Frobenius-norm weighting: $\rho_{\text{pooled}} = \sum_m w_m \rho_m / \sum_m w_m$, where $w_m = \|\Delta W_m\|_F$.

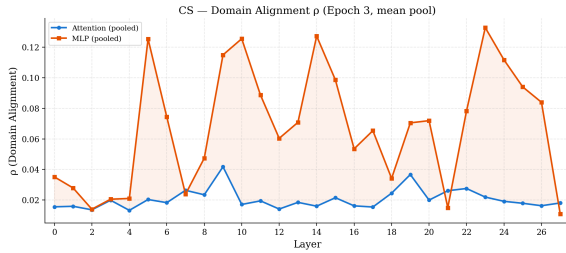
SVD overlap metric. Compute the SVD of ΔW and measure the total squared overlap of its top singular vectors with $\hat{d}_k^\ell$. This captures whether the principal directions of the weight update contain the domain concept direction, independent of magnitude.

5.3 Results

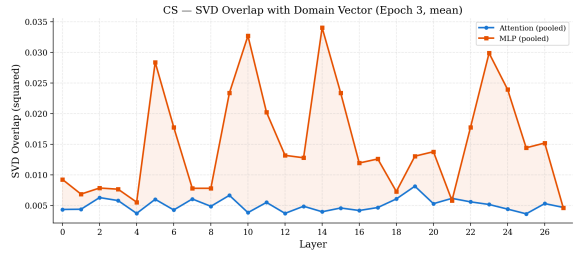
MLP updates show 2–4 $\times$ higher alignment with domain concept vectors than attention updates, consistently across all six domains.

Table 5: Mean domain alignment ρ across all layers (Epoch 3, mean pool). MLP exceeds attention by 2–4 $\times$ for every domain.

Domain	$\bar{\rho}_{\text{attn}}$	$\bar{\rho}_{\text{mlp}}$	Ratio
CS	0.0205	0.0678	3.31 $\times$
EESS	0.0205	0.0802	3.92 $\times$
Math	0.0174	0.0359	2.06 $\times$
Physics	0.0175	0.0554	3.16 $\times$
Q-Bio	0.0177	0.0411	2.32 $\times$
Stat	0.0188	0.0549	2.92 $\times$



(a) Projection ρ : CS



(b) SVD overlap: CS

Figure 6: Domain alignment across layers for CS (Epoch 3, mean pool). Left: projection metric ρ . Right: SVD overlap. In both, MLP (orange) dramatically exceeds attention (blue), with sharp peaks at layers 5, 10, 14, and 23.

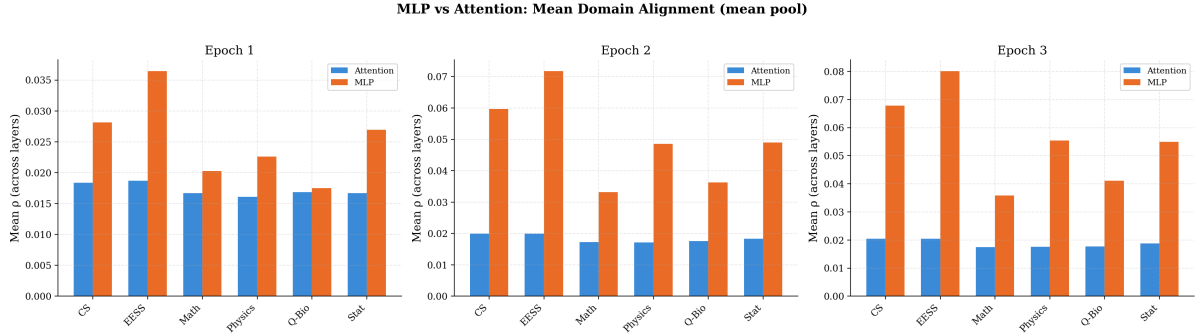


Figure 7: Summary: mean ρ across layers for all domains and all three epochs (mean pool). MLP consistently exceeds attention, and the gap *widens* with more training.

The alignment gap grows across epochs: at Epoch 1, the MLP/Attention ratio is $\sim 1.5\text{--}2\times$; by Epoch 3, it reaches $2\text{--}4\times$. This indicates that as the model trains longer, MLP updates become increasingly targeted toward domain-specific directions — even as the model begins to overfit on validation loss. The domain concept vectors act as a *directional signature* that MLP adapts toward.

6 Special Finding: The `gate_proj` Anomaly

Across all three experiments, the `gate_proj` matrix shows anomalously strong domain-specific behaviour, standing out from all other modules.

6.1 What is `gate_proj`?

In the SwiGLU MLP architecture used by Qwen2.5 (and Llama-family models), the MLP block at each layer computes:

$$\text{MLP}(x) = W_{\text{down}} \cdot [\sigma(W_{\text{gate}} \cdot x) \odot (W_{\text{up}} \cdot x)]$$

where σ is the SiLU activation and $\odot$ is element-wise multiplication. The `gate_proj` (W_{gate}) acts as a **feature amplifier**: it controls which intermediate features are activated via the gating mechanism. By selectively amplifying or suppressing features, it determines what information passes through the MLP. This makes it a natural locus for domain-specific adaptation.

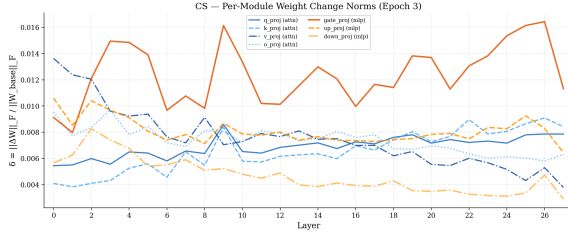
6.2 Evidence Across Experiments

Table 6: `gate_proj` vs. other modules (averaged across all domains, Epoch 3).

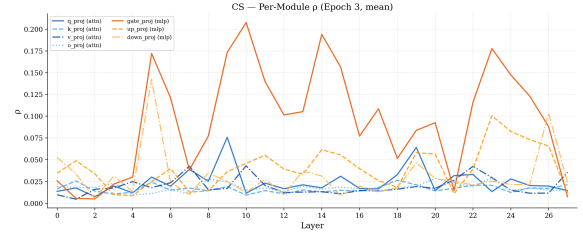
Module	$\bar{\delta}$ (weight norm)	$\bar{\rho}$ (alignment)	MSE mixing
q_proj	0.00685	0.0215	0.0107
k_proj	0.00658	0.0171	0.0145
v_proj	0.00765	0.0197	0.0902
o_proj	0.00744	0.0162	0.0431
gate_proj	0.01230	0.0767	0.0036
up_proj	0.00816	0.0338	0.0224
down_proj	0.00466	0.0297	0.0499

- **Weight change norm:** `gate_proj` has the highest δ of any module (0.0123), nearly $1.8\times$ the next-highest (up_proj at 0.0082) and $1.7\times$ the attention average.

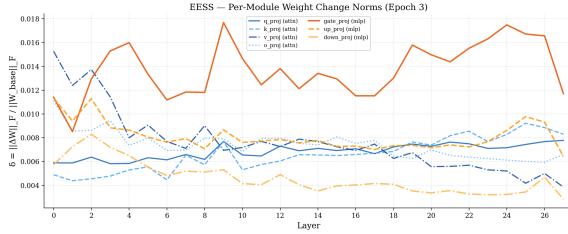
- **Domain alignment:** `gate_proj` has the highest ρ (0.0767), more than $2\times$ the next-highest MLP module (`up_proj` at 0.0338) and $3.8\times$ the attention average.
- **Domain specificity:** `gate_proj` has the *lowest* MSE mixing score (0.0036), meaning its updates are the most domain-differentiated of any module. It is $6\times$ more specialised than `up_proj`.



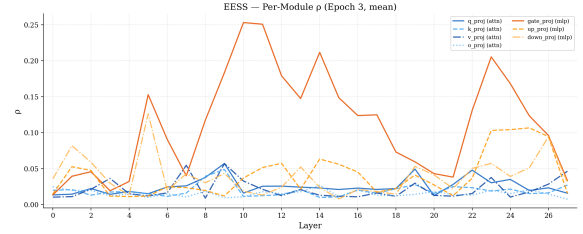
(a) Per-module weight change norms (CS).



(b) Per-module domain alignment ρ (CS).



(c) Per-module weight change norms (EES).



(d) Per-module domain alignment ρ (EES).

Figure 8: The `gate_proj` anomaly: in both weight change norms (left) and domain alignment (right), `gate_proj` (solid orange) dominates all other modules across layers. The pattern is consistent across domains.

Interpretation. The gating mechanism provides a natural explanation: `gate_proj` selects *which* MLP features are active, making it the first point of domain-specific routing within the MLP block. By modifying W_{gate} , fine-tuning can selectively activate domain-relevant features without changing the actual feature computation (W_{up}) or output projection (W_{down}) as much. This is consistent with the broader attention-router / MLP-executor framework: within the MLP itself, `gate_proj` acts as a *sub-router* that directs computation toward domain-specific feature subspaces.

7 Summary

Table 7: Summary of all three metrics (Epoch 3, averaged across domains).

Metric	Attention	MLP	MLP/Attn
Weight change norm ($\bar{\delta}$)	0.00699	0.00920	$1.31\times$
Domain mixing MSE (lower = specialised)	0.0275	0.0118	$2.33\times$
Domain alignment ρ (higher = aligned)	0.0189	0.0559	$2.96\times$

All three metrics consistently support the hypothesis: MLP modules adapt **more** ($1.3\times$ larger updates), **more specifically** ($2.3\times$ more domain-differentiated), and **more directionally** ($3.0\times$ more aligned with domain concept vectors) than attention modules. The `gate_proj`

matrix is the strongest individual contributor, acting as a domain-specific feature amplifier within the SwiGLU architecture.